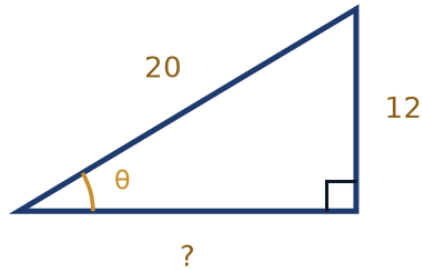


Trigonometry & Additional Topics



Right-triangle trigonometry

- 1 In the right triangle shown below, the hypotenuse is 20 and the side opposite angle θ is 12. Find the length of the adjacent side, then find $\sin \theta$, $\cos \theta$, and $\tan \theta$.



- 2 In a right triangle, angle θ has adjacent side 7 and hypotenuse 25. Find $\sin \theta$ and $\tan \theta$.
- 3 A 15-foot ladder leans against a wall, making a 50° angle with the ground. Find how high up the wall the ladder reaches, to the nearest tenth of a foot.

The Pythagorean identity

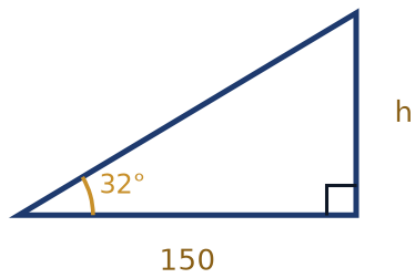
- 4 Given $\sin \theta = \frac{5}{13}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.
- 5 Given $\cos \theta = \frac{8}{17}$ with θ acute, find $\sin \theta$.

Exact trigonometric values

- 6 State the exact value of $\sin\left(\frac{\pi}{6}\right)$ and $\cos\left(\frac{\pi}{4}\right)$.
- 7 State the exact value of $\tan(60^\circ)$ and $\sin(90^\circ)$.

Solving for missing sides and angles

- 8 A surveyor stands 150 feet from the base of a tower and measures the angle of elevation to the top as 32° , as shown below. Find the height of the tower, to the nearest foot.



- 9 From the top of a 60-foot cliff, the angle of depression to a boat is 25° . Find the horizontal distance from the base of the cliff to the boat, to the nearest foot.

Complex numbers: arithmetic

- 10 Add $(4 + 3i) + (2 - 7i)$.
- 11 Multiply $(3 + 2i)(1 - 4i)$.
- 12 Simplify i^7 , using $i^2 = -1$.

Complex numbers: modulus and conjugates

- 13 Find the modulus $|6 - 8i|$.
- 14 Find the complex conjugate of $-3 + 5i$, and compute the product of the number and its conjugate.

Vectors: magnitude and operations

- 15 Vector $v = (3, 4)$. Find its magnitude.
- 16 Given vectors $u = (2, -5)$ and $v = (-6, 3)$, find $u + v$.
- 17 Given vector $w = (8, -6)$, find $2w$ and the magnitude of $2w$.

Radian measure

18 Convert 150° to radians, in terms of π .

19 Convert $\frac{7\pi}{4}$ radians to degrees.

A well-designed word problem: measuring the width of a river

20 This question has three parts. A surveyor stands at point A directly across a river from a tree at point B (so angle $A = 90^\circ$), then walks 80 meters along the riverbank to point C . (a) At point C , the angle between the bank and the line of sight to the tree is 35° . Use the tangent ratio to find AB , the width of the river, to the nearest meter. (b) Find the direct distance CB from C to the tree, to the nearest meter. (c) A colleague measures from a point 50 meters past C (so 130 meters from A) and finds the angle to the tree is 24° . Use this second measurement to find a second estimate of AB , to the nearest meter, and state which measurement gives the larger estimate.

Multiple choice — easy

21 In a right triangle, the hypotenuse is 87 and the side opposite angle θ is 60. Find $\sin \theta$.

- a. $\frac{21}{29}$
- b. $\frac{20}{29}$
- c. $\frac{20}{21}$
- d. $\frac{62}{87}$

22 In a right triangle, the side opposite angle θ is 6 and the side adjacent is 8. Find $\tan \theta$.

- a. 1
- b. $\frac{4}{5}$
- c. $\frac{3}{5}$
- d. $\frac{3}{4}$

23 In a right triangle, the hypotenuse is 50 and the side opposite angle θ is 14. Find $\sin \theta$.

a. $\frac{8}{25}$

b. $\frac{7}{25}$

c. $\frac{24}{25}$

d. $\frac{7}{24}$

24 In a right triangle, the side opposite angle θ is 24 and the side adjacent is 45. Find $\tan \theta$.

a. $\frac{8}{17}$

b. $\frac{15}{17}$

c. $\frac{26}{45}$

d. $\frac{8}{15}$

25 In a right triangle, the hypotenuse is 25 and the side adjacent to angle θ is 24. Find $\cos \theta$.

a. $\frac{26}{25}$

b. $\frac{24}{25}$

c. $\frac{7}{24}$

d. $\frac{7}{25}$

26 In a right triangle, the hypotenuse is 5 and the side opposite angle θ is 3. Find $\sin \theta$.

a. $\frac{4}{5}$

b. $\frac{3}{4}$

c. 1

d. $\frac{3}{5}$

- 27 In a right triangle, the side opposite angle θ is 40 and the side adjacent is 42. Find $\tan \theta$.
- a. 1
 - b. $\frac{21}{29}$
 - c. $\frac{20}{29}$
 - d. $\frac{20}{21}$
- 28 In a right triangle, the hypotenuse is 41 and the side adjacent to angle θ is 40. Find $\cos \theta$.
- a. $\frac{9}{40}$
 - b. $\frac{9}{41}$
 - c. $\frac{42}{41}$
 - d. $\frac{40}{41}$
- 29 In a right triangle, the hypotenuse is 15 and the side opposite angle θ is 9. Find $\sin \theta$.
- a. $\frac{3}{4}$
 - b. $\frac{11}{15}$
 - c. $\frac{4}{5}$
 - d. $\frac{3}{5}$
- 30 In a right triangle, the side opposite angle θ is 18 and the side adjacent is 80. Find $\tan \theta$.
- a. $\frac{40}{41}$
 - b. $\frac{9}{41}$
 - c. $\frac{9}{40}$
 - d. $\frac{1}{4}$

- 31 In a right triangle, the hypotenuse is 26 and the side adjacent to angle θ is 24. Find $\cos \theta$.
- a. $\frac{5}{12}$
 - b. 1
 - c. $\frac{12}{13}$
 - d. $\frac{5}{13}$
- 32 In a right triangle, the hypotenuse is 51 and the side opposite angle θ is 24. Find $\sin \theta$.
- a. $\frac{15}{17}$
 - b. $\frac{26}{51}$
 - c. $\frac{8}{15}$
 - d. $\frac{8}{17}$
- 33 In a right triangle, the hypotenuse is 29 and the side opposite angle θ is 20. Find $\sin \theta$.
- a. $\frac{21}{29}$
 - b. $\frac{20}{29}$
 - c. $\frac{22}{29}$
 - d. $\frac{20}{21}$
- 34 In a right triangle, the hypotenuse is 10 and the side adjacent to angle θ is 8. Find $\cos \theta$.
- a. 1
 - b. $\frac{4}{5}$
 - c. $\frac{3}{4}$
 - d. $\frac{3}{5}$

35 In a right triangle, the side opposite angle θ is 36 and the side adjacent is 105. Find $\tan \theta$.

- a. $\frac{12}{35}$
- b. $\frac{38}{105}$
- c. $\frac{12}{37}$
- d. $\frac{35}{37}$

Multiple choice — medium

36 Given $\sin \theta = \frac{11}{61}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

- a. $\frac{62}{61}$
- b. $\frac{60}{61}$
- c. 1
- d. $\frac{11}{61}$

37 Given $\sin \theta = \frac{12}{37}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

- a. $\frac{35}{36}$
- b. $\frac{12}{37}$
- c. $\frac{36}{37}$
- d. $\frac{35}{37}$

38 Given $\sin \theta = \frac{9}{41}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.

- a. $\frac{40}{41}$
- b. $\frac{9}{41}$
- c. 1
- d. $\frac{42}{41}$

- 39 Given $\sin \theta = \frac{5}{13}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.
- a. $\frac{14}{13}$
 - b. 1
 - c. $\frac{12}{13}$
 - d. $\frac{5}{13}$
- 40 Given $\cos \theta = \frac{21}{29}$ with θ acute, find $\sin \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.
- a. $\frac{20}{29}$
 - b. $\frac{22}{29}$
 - c. $\frac{21}{29}$
 - d. $\frac{5}{7}$
- 41 Given $\cos \theta = \frac{15}{17}$ with θ acute, find $\sin \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.
- a. $\frac{9}{17}$
 - b. $\frac{1}{2}$
 - c. $\frac{8}{17}$
 - d. $\frac{15}{17}$
- 42 Given $\cos \theta = \frac{60}{61}$ with θ acute, find $\sin \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.
- a. $\frac{11}{60}$
 - b. $\frac{60}{61}$
 - c. $\frac{12}{61}$
 - d. $\frac{11}{61}$

- 43 Given $\sin \theta = \frac{7}{25}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.
- a. $\frac{7}{25}$
 - b. $\frac{26}{25}$
 - c. 1
 - d. $\frac{24}{25}$
- 44 Given $\cos \theta = \frac{35}{37}$ with θ acute, find $\sin \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.
- a. $\frac{12}{37}$
 - b. $\frac{35}{37}$
 - c. $\frac{13}{37}$
 - d. $\frac{1}{3}$
- 45 Given $\cos \theta = \frac{4}{5}$ with θ acute, find $\sin \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.
- a. $\frac{4}{5}$
 - b. 1
 - c. $\frac{3}{5}$
 - d. $\frac{3}{4}$
- 46 Given $\sin \theta = \frac{20}{29}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.
- a. $\frac{22}{29}$
 - b. $\frac{20}{29}$
 - c. $\frac{21}{29}$
 - d. $\frac{3}{4}$

- 47 Given $\cos \theta = \frac{12}{13}$ with θ acute, find $\sin \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.
- a. $\frac{12}{13}$
 - b. $\frac{5}{12}$
 - c. $\frac{5}{13}$
 - d. $\frac{6}{13}$
- 48 Given $\sin \theta = \frac{3}{5}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.
- a. $\frac{4}{5}$
 - b. $\frac{3}{5}$
 - c. $\frac{6}{5}$
 - d. 1
- 49 Given $\cos \theta = \frac{40}{41}$ with θ acute, find $\sin \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.
- a. $\frac{10}{41}$
 - b. $\frac{40}{41}$
 - c. $\frac{9}{41}$
 - d. $\frac{9}{40}$
- 50 Given $\cos \theta = \frac{24}{25}$ with θ acute, find $\sin \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.
- a. $\frac{8}{25}$
 - b. $\frac{24}{25}$
 - c. $\frac{7}{25}$
 - d. $\frac{7}{24}$

Multiple choice — hard

51 Multiply $(8 - i)(1 + 5i)$.

a. $16 + 39i$

b. $13 + 39i$

c. $3 + 39i$

d. $13 + 41i$

52 Multiply $(1 - 3i)(9 - 9i)$.

a. $-18 + 18i$

b. $-18 - 34i$

c. $36 - 36i$

d. $-18 - 36i$

53 Multiply $(-1 - 9i)(5 + 2i)$.

a. $13 - 47i$

b. $-23 - 47i$

c. $13 + 43i$

d. $13 - 45i$

54 Multiply $(6 - 6i)(-2 - 8i)$.

a. $-60 - 36i$

b. $36 - 36i$

c. $-60 - 34i$

d. $-60 - 60i$

- 55 Multiply $(7 + 3i)(-1 - 8i)$.
- a. $-31 - 59i$
 - b. $17 - 59i$
 - c. $17 - 53i$
 - d. $17 - 57i$
- 56 Given vectors $u = (0, 5)$ and $v = (7, 0)$, find $u + v$.
- a. $(7, 3)$
 - b. $(9, 5)$
 - c. $(5, 7)$
 - d. $(7, 5)$
- 57 Given vectors $u = (9, -8)$ and $v = (5, -3)$, find $u + v$.
- a. $(-11, 14)$
 - b. $(14, -11)$
 - c. $(14, -13)$
 - d. $(16, -11)$
- 58 Given vectors $u = (4, 9)$ and $v = (1, -7)$, find $u + v$.
- a. $(2, 5)$
 - b. $(5, 2)$
 - c. $(5, 0)$
 - d. $(7, 2)$

59 Given vectors $u = (3, -1)$ and $v = (-8, -5)$, find $u + v$.

a. $(-5, -6)$

b. $(-3, -6)$

c. $(-5, -8)$

d. $(-6, -5)$

60 Given vectors $u = (-4, -2)$ and $v = (0, 5)$, find $u + v$.

a. $(-4, 1)$

b. $(3, -4)$

c. $(-2, 3)$

d. $(-4, 3)$

61 Convert 90° to radians, in terms of π .

a. π

b. $\frac{3\pi}{2}$

c. $\frac{\pi}{3}$

d. $\frac{\pi}{2}$

62 Convert 120° to radians, in terms of π .

a. $\frac{4\pi}{3}$

b. $\frac{\pi}{2}$

c. $\frac{2\pi}{3}$

d. π

63 Convert 75° to radians, in terms of π .

a. $\frac{5\pi}{6}$

b. $\frac{\pi}{2}$

c. $\frac{5\pi}{13}$

d. $\frac{5\pi}{12}$

64 Convert 345° to radians, in terms of π .

a. $\frac{23\pi}{13}$

b. $\frac{23\pi}{6}$

c. $\frac{23\pi}{12}$

d. 2π

65 Convert 180° to radians, in terms of π .

a. π

b. 3π

c. $\frac{\pi}{2}$

d. 2π